Introduction to Artificial Intelligence and Machine Learning (CSN11601)

K-Mean Clustering

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A. Marketing Strategy

You have identified clusters of customers. For each cluster, propose a targeted marketing strategy. Which cluster would be most appropriate for premium product targeting, and why?

Sample Input:

CustomerID	Age	Annual Income (k$)	Spending Score (1–100)	Cluster
12	26	78	85	2
45	41	120	40	3

Sample Output:

Cluster 2 – Young, high-spending customers: Target with luxury product ads and exclusive deals.
Cluster 3 – Middle-aged, high-income, low-spending: Offer premium subscription or loyalty benefits.

B. Product/Service Design

Your company plans to launch a new loyalty program or customized service. Based on the cluster profiles, identify which clusters should be prioritized for this initiative and design at least two personalized offers or services per cluster.

Sample Input:

Cluster Summary:
- Cluster 0: Low Income – Low Spending
- Cluster 1: High Income – High Spending
- Cluster 2: Middle Income – Moderate Spending

Sample Output:

- Cluster 1 (High Income – High Spending):
- Personalized luxury offers (early access to new products)
- VIP membership program with cashback rewards
- Cluster 2 (Middle Income – Moderate Spending):
- Discount coupons for repeat purchases
- Loyalty points redeemable for free delivery

A1. CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

df = pd.read_csv('Mall_Customers.csv')

data = df.drop(columns=['CustomerID'])
data['Gender'] = data['Gender'].map({'Male': 0, 'Female': 1})

scaler = StandardScaler()
scaled_data = scaler.fit_transform(data[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']])
scaled_df = pd.DataFrame(scaled_data, columns=['Age', 'Annual Income (k$)', 'Spending Score (1-100)'])

wcss = []
K = range(1, 11)
for k in K:
    model = KMeans(n_clusters=k, random_state=42)
    model.fit(scaled_df)
    wcss.append(model.inertia_)

plt.figure(figsize=(8, 5))
plt.plot(K, wcss, 'bo-')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('WCSS')
plt.title('Elbow Method for Optimal k')
plt.savefig('elbow_plot.png')
plt.close()
print("Elbow plot saved as: elbow_plot.png")

kmeans = KMeans(n_clusters=5, random_state=42)
kmeans.fit(scaled_df)

scaled_df['Cluster'] = kmeans.labels_
data['Cluster'] = kmeans.labels_

plt.figure(figsize=(10, 7))
sns.scatterplot(
    data=scaled_df,
    x='Annual Income (k$)',
    y='Spending Score (1-100)',
    hue='Cluster',
    palette='Set2',
    s=80
)
plt.title('Customer Segments based on Income and Spending')
plt.savefig('cluster_plot.png')
plt.close()
print("Cluster plot saved as: cluster_plot.png")

score = silhouette_score(scaled_df.drop(columns=['Cluster']), scaled_df['Cluster'])
print(f"Silhouette Score: {score:.3f}")

cluster_summary = data.groupby('Cluster')[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']].mean()
print("\nCluster Summary:")
print(cluster_summary)
```

OUTPUT:

```
(venv) aayansh207@aayansh-PC:~/Desktop/MNNIT/Cluster$ python3 p1.py
Elbow plot saved as: elbow_plot.png
Cluster plot saved as: cluster_plot.png
Silhouette Score: 0.408

Cluster Summary:
                Age  Annual Income (k$)  Spending Score (1-100)
Cluster
0         55.275862           47.620690               41.706897
1         32.875000           86.100000               81.525000
2         25.769231           26.115385               74.846154
3         26.733333           54.311111               40.911111
4         44.387097           89.774194               18.483871
```



